CS315 · ALGORITHMS ANALYSIS & DESIGN

Practice 3 From the 461 Final

Only the 461 final questions that fall within the midterm

Muharib

01 Before you start

These questions are taken from the 461 final (26/12/2024). I kept only the questions whose topics fall within the midterm, and left out the questions on graphs, Kruskal, Dijkstra, Bellman-Ford and the Fibonacci heap because they come after the midterm.

I have no answer key for the final, so the solutions are my own. They are in a separate file.

02 Questions

Q1 MCQ · FINAL Q1

3) If several elements are competing for the same bucket or position in the hash table, what is it called?

a. Diffusion b. Replication c. Collision d. Duplication

4) Which of the following is not a technique to avoid a collision?

a. Make the hash function appear random

b. Use the chaining method

c. Use uniform hashing

d. Increasing hash table size

Q2 SHORT ANSWERS · FINAL Q2 · 8 MARKS

A. What are the three steps of Divide and conquer approach

B. Give two examples of Sorting algorithms using the Divide and conquer approach

C. If we have a B-tree in the order of 7 find:

1\ Max number of child = _____

2\ Max number of key = _____

3\ Min number of key = _____

Q3 ANALYSIS · FINAL Q3 C

Analyse (in details) the following algorithm Theoretically

```
Algorithm F1(a, n)
1. w ← 0
2. c ← 1
3. while w < n² do
4. w ← w + 1
5. c ← c * a
6. return c
```

Q4 B-TREE · FINAL Q4 C

Assume you have an empty B-tree of degree 3.
Draw (in details) the insert operation for the following elements: 8, 9, 10, 11, 15, 20.